Pocket Arithmetic and Interaction in WPM

A mathematical study on signed responses, represented fields and relational operators

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Abstract

This paper develops a conditional calculus for manipulating individual pockets and their represented contributions. It separates an extracted basin, a signed response packet, an additive field component and a physical source. Starting from aligned linear probes, it derives interference identities, exact partition arithmetic, Gaussian kernel operations, overlap matrices and full-grid interaction fields. Rank conditions characterize identifiable relational coefficients, while perturbation bounds explain the consequences of approximation. Worked equation trails and exercises provide a reproducible application procedure.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

1 Learning objectives and the object being manipulated

The prerequisite material is matrix algebra, differentiation, basic probability and least squares. The central question is precise: *what object is being added, multiplied or removed?* A pocket is a connected region of a low-risk landscape. A signed packet is a collection of oriented measurements or errors. A field is a function on declared coordinates. A physical source is a hypothesized part of a data-generating mechanism. None of these definitions makes the four objects interchangeable.

Let t denote the issuance time, $P > 0$ the requested horizon, $W > 0$ the historical extent and $m > 0$ a declared sampling or internal-resolution parameter. Write $r = (W, m)$ for a resolution

coordinate; keep different targets, horizons, channels and regimes in separate layers unless an explicit map aligns them. A pocket record may be written

$$\mathcal{P}_i(t) = (id_i, \Omega_i, z_i, e_i, S_i, H_i, M_i). \tag{1.1}$$

Here Ω_i is support, z_i contains location and geometry, e_i denotes detected existence, S_i is signed payload, H_i is verified history and M_i stores units, orientation, noise and provenance. A flow curve means a coordinate-tagged cross-section through the landscape, such as $W \mapsto L_t(W, m_i)$; it is not automatically a velocity.

Definition 1.1 (Aligned arithmetic)**.** Two operands are aligned when their time coordinates, channel meanings, measurement units, domain masks, probe rows and normalization conventions agree. If a remapping is necessary, that remapping is part of the operation and its error must be retained.

For example, adding two displacements measured at the same positions can be meaningful. Adding their prediction-error variances generally requires covariance terms. Concatenating two groups of observations is a third operation. Naming all three "adding datasets" conceals different mathematics.

2 Linear probes and the signed carrier

Let $X \in \mathbb{R}^{n\times d}$ contain n observed times and d channels. A fixed linear operator $Q \in \mathbb{R}^{q\times n}$ produces $S = QX$. A forecast-error row can have the form $q_s^\top = e_{s+P}^\top - a_s^\top R_s$, where R_s selects the past window and a_s contains forecast coefficients frozen at historical issuance s. The target must have matured: $s + P \le t$. This fixes the sign as observed target minus prediction.

Choose a positive semidefinite channel metric G and row weights $D \succeq 0$. With positive constant c, define

$$\langle S_A, S_B\rangle_{D,G} = c^{-1}\,\mathrm{tr}(S_A^\top D S_B G), \qquad U(S) = \langle S, S\rangle_{D,G}\,. \tag{2.1}$$

The quantity is a seminorm if either metric is singular. Equal weighting uses $D = I$, $G = I$, $c = q$. An energy landscape $r \mapsto U(Q_r X)$ is a particular risk construction, not the definition of every WPM landscape.

Theorem 2.1 (Signed superposition)**.** *For fixed aligned Q, D, G and real a, b,*

$$S_{aA+bB} = aS_A + bS_B, \tag{2.2}$$

$$U_{aA+bB} = a^2U_A + b^2U_B + 2ab\Gamma_{AB}, \qquad \Gamma_{AB} = \langle S_A, S_B\rangle_{D,G}\,. \tag{2.3}$$

Proof. Linearity gives $Q(aA + bB) = aQA + bQB$. Substitute into the bilinear form and expand. Symmetry of D and G makes the two cross terms equal. The calculation does not apply to an operator that is refitted differently on the sum unless the refitting map itself has the required linearity. □

For K sources the same expansion is $U = \sum_i a_i^2 U_i + 2\sum_{i<j} a_i a_j \Gamma_{ij}$. The matrix (Γ_{ij}) is positive semidefinite because $a^\top \Gamma a = U(\sum_i a_i S_i) \ge 0$. This is a useful consistency check for stored pair summaries.

Corollary 2.2 (What magnitudes alone permit)**.** *For $a = b = 1$,*

$$(\sqrt{U_A} - \sqrt{U_B})^2 \le U_{A+B} \le (\sqrt{U_A} + \sqrt{U_B})^2. \tag{2.4}$$

Proof. Cauchy–Schwarz for the weighted seminorm gives $|\Gamma_{AB}| \le \sqrt{U_A U_B}$. Insert both endpoints into the superposition identity. □

Worked example 2.3 (A complete cancellation trail)**.** Take $S_A = (1, 2)$, $S_B = (-1, -2)$ and $c = 2$. Then $U_A = U_B = 5/2$ and $\Gamma = -5/2$. Thus $U_{A+B} = 5 - 5 = 0$. Replacing S_B by $(1, 2)$ leaves both individual energies unchanged but yields $U_{A+B} = 10$. Scalar pocket depth cannot distinguish these cases.

Application rule. Preserve signed rows when cancellation matters. A positive geometric overlap says nothing by itself about the sign of Γ, and $\Gamma \neq 0$ does not establish a causal interaction between physical sources.

3 Pooling, channel stacking and dependence

For disjoint observation groups and the same fixed predictor, loss and weighting convention,

$$L_{\text{pool}} = \frac{n_A L_A + n_B L_B}{n_A + n_B}. \tag{3.1}$$

This identity concerns averaging losses. It contains no interference term because there is no pointwise signal sum. A refitted pooled predictor has a new risk and must be evaluated as such. Channel stacking produces a larger observation vector; it requires a joint noise covariance and a meaningful channel metric.

If $\mathrm{vec}(X)$ has covariance Σ_X, vectorization gives

$$\Sigma_S = (I_d \otimes Q)\Sigma_X(I_d \otimes Q)^\top. \tag{3.2}$$

Overlapping probe rows are usually correlated even when raw errors are independent. Repeatedly storing a row cannot create information. With two identical rows and scalar noise variance σ^2, the row covariance is $\sigma^2 \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, of rank one. Treating it as diagonal falsely invents a second independent measurement.

4 An exact additive representation for field operations

Let $V : \mathcal{X} \to \mathbb{R}$ be a declared carrier field with measure μ. To manipulate components, choose

$$V(x) = b(x) + \sum_{i=1}^{K} a_i \psi_i(x) + R(x). \tag{4.1}$$

The background b, basis functions ψ_i, coefficients a_i and residual R are all part of the representation. A negative a_i may represent a depression. This does not imply a negative observation residual or inhibitory physical coupling.

Proposition 4.1 (Exact partition with residual)**.** *Let measurable ownership functions satisfy* $0 \le \chi_i(x)$ *and* $\sum_i \chi_i(x) \le 1$. *Define* $V_i = \chi_i(V - b)$ *and* $R = (1 - \sum_i \chi_i)(V - b)$. *Then* $V = b + \sum_i V_i + R$ *pointwise.*

Proof. The coefficient of $V - b$ in the right-hand side is $\sum_i \chi_i + 1 - \sum_i \chi_i = 1$. This proves reconstruction, but not uniqueness of ownership or physical source attribution. □

Addition of a represented contribution is $V' = V + a_*\psi_*$. Removal is $V' = V - a_k\psi_k$. Translation by δ is

$$V' = V - a_k\psi_k + a_k\psi_k(\,\cdot - \delta\,). \tag{4.2}$$

The new field must be segmented again. Two components may merge into one basin or erase an existing basin. Hence there is no general formula "number of new pockets equals the sum of old pocket counts." Collection union, which merely preserves records and provenance, is not field superposition.

On bounded domains, translation requires a boundary rule: truncation, reflection or periodic wrapping. Truncation can lose integrated mass. A periodic extension assumes that opposite boundaries are neighbors. Neither is a harmless display convention.

5 Gaussian kernels: product, convolution and addition

For positive definite $d \times d$ covariance matrices, define unit-height kernels

$$\psi_A(x) = \exp[-\tfrac{1}{2}(x-\mu_A)^\top \Sigma_A^{-1}(x-\mu_A)]. \tag{5.1}$$

These are not normalized probability densities. Amplitudes and determinant factors must therefore be carried explicitly.

Theorem 5.1 (Pointwise Gaussian product)**.** *Set* $\Sigma_C = (\Sigma_A^{-1} + \Sigma_B^{-1})^{-1}$ *and* $\mu_C = \Sigma_C(\Sigma_A^{-1}\mu_A + \Sigma_B^{-1}\mu_B)$*. Then*

$$\psi_A(x)\psi_B(x) = \kappa \exp[-\tfrac{1}{2}(x-\mu_C)^\top \Sigma_C^{-1}(x-\mu_C)], \tag{5.2}$$

$$\kappa = \exp[-\tfrac{1}{2}(\mu_A-\mu_B)^\top (\Sigma_A+\Sigma_B)^{-1}(\mu_A-\mu_B)]. \tag{5.3}$$

Proof. Expand both quadratic forms. Their quadratic coefficient is $\Sigma_A^{-1} + \Sigma_B^{-1}$ and their linear coefficient is $-2(\Sigma_A^{-1}\mu_A + \Sigma_B^{-1}\mu_B)$. Complete the square. The remaining constant is the displayed displacement quadratic, as follows either by substituting μ_C or by the matrix parallel-sum identity. □

Consequently the integral overlap is $\kappa(2\pi)^{d/2}|\Sigma_C|^{1/2}$. On all of $\mathbb{R}^d$, convolution instead gives

$$(\psi_A * \psi_B)(x) = (2\pi)^{d/2}\frac{|\Sigma_A|^{1/2}|\Sigma_B|^{1/2}}{|\Sigma_A+\Sigma_B|^{1/2}} e^{-\frac{1}{2}(x-\mu_A-\mu_B)^\top(\Sigma_A+\Sigma_B)^{-1}(x-\mu_A-\mu_B)}. \tag{5.4}$$

To derive this, write each unit-height kernel as $(2\pi)^{d/2}|\Sigma|^{1/2}$ times its normalized density and use the Gaussian sum density, or complete the square inside the convolution integral. Multiplication narrows; convolution broadens; addition usually preserves two modes or a non-Gaussian shape.

Worked example 5.2 (Three different operations)**.** In one dimension let $(\mu_A, s_A^2) = (0, 1)$ and $(\mu_B, s_B^2) = (1, 1/4)$. The product has center $4/5$, variance $1/5$ and height multiplier $e^{-2/5}$. Its overlap integral is $e^{-2/5}\sqrt{2\pi/5}$. Convolution has center 1, variance $5/4$ and height multiplier $\sqrt{2\pi/5}$. The sum is simply $e^{-x^2/2} + e^{-2(x-1)^2}$, not either result.

Application rule. Use Gaussian algebra only for declared Gaussian components. A fitted Gaussian basin is an approximation to a landscape, and its residual must not disappear from subsequent calculations.

6 Overlap matrices and spatial interaction fields

On a grid x_c with positive cell volumes v_c, let $P_{A,ic} = \psi_{A,i}(x_c)$ and $P_{B,jc} = \psi_{B,j}(x_c)$. Define

$$O_{ij} = \sum_c v_c P_{A,ic} P_{B,jc}, \tag{6.1}$$

$$J_c = \sum_{i,j} K_{ij} P_{A,ic} P_{B,jc}. \tag{6.2}$$

O integrates away position; J retains it. A normalized geometric overlap is $C_{ij} = O_{ij}/(\|P_{A,i}\|_v \|P_{B,j}\|_v)$ when both norms are nonzero. For nonnegative memberships, $0 \le C_{ij} \le 1$. Binary-mask intersection-over-union, Dice overlap and this cosine overlap use different denominators.

Worked example 6.1 (Recovering a full interaction field)**.** Take unit cell volumes, $P_{A,1} = (1,0,0,0)$, $P_{A,2} = (0,0,2,0)$ and $P_{B,1} = (1/2,1,1,0)$. Then $O = (1/2,2)^\top$. With $K_{11} = 2$ and $K_{21} = -1/2$, equation (6.2) gives $J = (1,0,-1,0)$. The overlap vector alone cannot tell which cells receive these signs.

Proposition 6.2 (Label invariance and spatial information loss)**.** *Permuting pocket labels and the corresponding indices of K leaves J unchanged. The map from pair-product fields to O is generally noninjective when at least two cells are available.*

Proof. The first claim is a relabeling of finite summation indices. For the second, any nonzero cell vector h with $\sum_c v_c h_c = 0$ can be added to a pair field without changing its integral. More simply, moving a unit product from one equal-volume cell to another preserves its overlap value while changing location. □

Store P_A, P_B or their full pair fields if later spatial reconstruction is required. Ordinary matrix multiplication of two image arrays sums over an index; it is not the pointwise product and has no spatial-interaction meaning unless the intermediate index has been defined.

7 Division, differentiation and integration of carriers

Quotients are allowed only on a declared domain where the denominator is nonzero. If the denominator is an estimated weak response, inversion can amplify measurement error. A small risk value does not authorize dividing one physical field by another unless that quotient is the intended mathematical target.

Proposition 7.1 (A quotient perturbation bound)**.** *Assume $|B(x)|, |\widehat{B}(x)| \ge b_0 > 0$ on the retained domain, $\left\|A - \widehat{A}\right\|_\infty \le \varepsilon_A$ and $\left\|B - \widehat{B}\right\|_\infty \le \varepsilon_B$. Then*

$$\left\| \frac{A}{B} - \frac{\widehat{A}}{\widehat{B}} \right\|_\infty \le \frac{\varepsilon_A}{b_0} + \frac{\left\|\widehat{A}\right\|_\infty \varepsilon_B}{b_0^2}. \tag{7.1}$$

Proof. Subtract the quotients as $(A - \widehat{A})/B + \widehat{A}(\widehat{B} - B)/(B\widehat{B})$ and apply the lower denominator bound and triangle inequality. □

If only $|\widehat{B}| \ge \widehat{b}$ is known, a bound $\varepsilon_B < \widehat{b}$ guarantees $|B| \ge \widehat{b} - \varepsilon_B$. Otherwise the true denominator may cross zero. Return a masked or branch-sensitive result rather than silently clipping the denominator and calling the quotient exact. Clipping defines a different operator.

For differentiable carriers, differentiation is linear and obeys the ordinary product and quotient rules. It acts on the declared coordinate x, which may be physical position or a resolution coordinate. Differentiating a risk surface with respect to W does not produce physical velocity. Likewise, $\int V\,d\mu = \int b\,d\mu + \sum_i a_i \int \psi_i\,d\mu + \int R\,d\mu$ is a linear carrier summary, not necessarily a normalized probability.

Discrete derivatives deserve their own algebra. With $D_h A(x) = [A(x+h) - A(x)]/h$,

$$D_h(AB)(x) = A(x+h)D_hB(x) + B(x)D_hA(x). \tag{7.2}$$

The proof expands the right-hand side and cancels the intermediate $A(x+h)B(x)$ terms. Using only A(x) and B(x) in an unshifted continuous product formula omits a term $hD_hA(x)D_hB(x)$. Thus an identity proved for smooth continuous fields is not automatically exact for a chosen discrete operator. Store the grid and derivative convention with the result.

8 Learning a relational operator

Equation (6.2) specifies a basis, not a universal law. If observed mixed fields $Y^{(e)}$ and baselines $b^{(e)}$ are available, stack their pair-product columns into a design Z and fit

$$y - b = Z\theta + \epsilon, \qquad \widehat{\theta} = \arg\min_{\theta} (y - b - Z\theta)^{\top}\Sigma^{-1}(y - b - Z\theta) + \lambda \left\|\theta\right\|^2. \tag{8.1}$$

Here θ vectorizes the allowed K entries. Mask incompatible pairs before fitting. If Σ is singular, work on its supported subspace. A noiseless covariance-null direction is an exact constraint if the model asserts zero variance there; it must not simply be discarded.

Theorem 8.1 (Identifiable coefficients)**.** *For a known baseline, fixed design and noiseless data, θ is unique exactly when* $\ker Z = \{0\}$*. With nuisance columns D and unconstrained nuisance coefficients, uniqueness of θ requires* $\operatorname{rank}[(I - DD^{+})Z] = \dim\theta$ *after any invertible whitening.*

Proof. Two solutions differ by $Z\delta = 0$. With nuisance effects, $Z\delta$ can be canceled by $D\gamma$ exactly when its projection orthogonal to the columns of D is zero. The projected design must therefore have a trivial nullspace. □

Ridge makes an optimizer unique even when this rank condition fails; it does not make the physical coefficient unique. For independent correctly specified Gaussian errors and full rank, the unregularized covariance is $(Z^{\top}\Sigma^{-1}Z)^{-1}$. Errors in P_A, P_B create a noisy design and invalidate that formula unless their uncertainty is modeled.

9 Approximation, compression and error propagation

Suppose $A = \widehat{A} + e_A$, $B = \widehat{B} + e_B$, and $\left\|e_A\right\|_\infty \le \varepsilon_A$, $\left\|e_B\right\|_\infty \le \varepsilon_B$. Then

$$\left\|(A+B) - (\widehat{A} + \widehat{B})\right\|_\infty \le \varepsilon_A + \varepsilon_B, \tag{9.1}$$

$$\left\|AB - \widehat{A}\widehat{B}\right\|_\infty \le \left\|\widehat{A}\right\|_\infty \varepsilon_B + \left\|\widehat{B}\right\|_\infty \varepsilon_A + \varepsilon_A\varepsilon_B. \tag{9.2}$$

The proof is expansion of $(\widehat{A} + e_A)(\widehat{B} + e_B)$ followed by the triangle inequality. Small amplitude error need not preserve shallow-basin topology. Thus field error and pocket-existence error are distinct quantities.

For a fixed least-squares dictionary $Z = UR$ with $U^\top U = I$, orthogonal decomposition yields

$$\|y - Z\theta\|^2 = \left\|U^\top y - R\theta\right\|^2 + \left\|(I - UU^\top)y\right\|^2. \tag{9.3}$$

The final term is constant in θ. Keeping $R, U^\top y$ and the residual norm preserves the objective for that dictionary. It does not preserve arbitrary fields or support later fitting to columns outside its span. Standard least-squares and optimization references are [3, 4].

10 A complete application procedure

1. Specify the carrier, coordinates, measure, target and information cutoff. Name the operation: signal sum, loss pooling, component edit or relational fitting.
2. Align units, channel orientation, masks and operator versions. Retain raw anchors for directions annihilated by probes.
3. Preserve pocket flow slices, position, existence and individual identities; retain background and residual fields.
4. Execute the appropriate exact formula. For interaction coefficients, declare a dictionary and inspect data rank and conditioning.
5. Propagate noise and approximation errors. Distinguish algebraic equality on stored arrays from a claim about unobserved physical detail.
6. Re-extract basins after field changes. Keep split, merge and uncertain-lineage records rather than forcing one persistent label.
7. If a forecast is requested, attach a known or learned predictor. Arithmetic correctness alone does not determine future operands or future coefficients.

Do not add risk heights as signal amplitudes, replace an overlap matrix by a spatial map, infer interaction sign from geometric overlap, or call a representation edit a physical intervention.

11 A complete relational arithmetic worksheet

Consider two aligned binary inputs A and B and an observed scalar carrier

$$Y = \mu + \alpha A + \beta B + \gamma AB. \tag{11.1}$$

The term AB is a declared interaction basis. Four observations, one at each input pair, are

A	B	Y	AB
0	0	1	0
1	0	3	0
0	1	4	0
1	1	10	1

The design rows are $(1, A, B, AB)$. Instead of invoking a black-box inverse, solve them in order:

$$\mu = Y_{00} = 1, \tag{11.2}$$

$$\alpha = Y_{10} - Y_{00} = 2, \qquad \beta = Y_{01} - Y_{00} = 3, \tag{11.3}$$

$$\gamma = Y_{11} - Y_{10} - Y_{01} + Y_{00} = 4. \tag{11.4}$$

This is a contrast in a full-rank model. If one observed only $A = B$, columns A and B would coincide and AB would also equal A at binary values; three effects would be confounded. More precise copies at those same two settings would not identify them separately.

Suppose instead the four Y observations have independent equal-variance errors of variance σ^2. The interaction contrast uses coefficients $(1, -1, -1, 1)$ and has variance $4\sigma^2$. With arbitrary covariance Σ across these observations it has variance $c^\top \Sigma c$. A contrast is not four independent new observations; it is one derived response with a calculable covariance. With M genuine independent readings at each setting and no common error, the variance becomes $4\sigma^2/M$.

For a spatial version, let Y_c satisfy the same equation at every grid cell c. Then $\gamma_c = Y_{11,c} - Y_{10,c} - Y_{01,c} + Y_{00,c}$ recovers a full-grid interaction field under the shared model. Integrating first recovers only the corresponding integrated interaction and cannot locate its support. Smoothness penalties can stabilize a noisy γ_c field, but they change the estimator and impose an assumption about spatial variation.

To connect this worksheet to actual pockets, build causal risk surfaces for forecasts of each measured carrier, extract pockets, and retain the signed contrasts associated with their lenses. The four factorial settings are not automatically four pockets. Their role is to identify a relational coefficient through independently varied inputs. The pocket representation can preserve where that relation is useful without redefining the underlying algebra.

12 Products as mechanisms, products as evidence

The same multiplication sign can encode three different constructions. In $Y = A + B + \gamma AB$, it is a candidate physical or statistical mechanism. In $\psi_A \psi_B$, it is an overlap basis on a spatial domain. In $p(D_1 \mid \theta)p(D_2 \mid \theta)$, it is a likelihood factorization requiring the stated conditional independence or an equivalent joint model. None licenses the others.

For example, two noisy sensors observing the same latent field may have correlated errors. Their likelihood is a joint density with full covariance, not automatically a product of their marginal densities. Similarly, multiplying two pocket probability densities and renormalizing is a product-of-experts model; it is not the probability distribution of adding their physical fields. Mixture combination $\sum_i \pi_i p_i$ expresses an alternative model label, whereas convolution can describe a sum of independent random variables. The mathematical operation follows the intended mechanism.

The Gaussian example makes the distinction visible.

13 Dimension checks and a reusable arithmetic record

Every formula should pass a unit check. If A and B are displacements, AB has squared-displacement units, so a coefficient multiplying AB into a displacement equation has inverse-displacement units. An overlap integral has kernel-product units times domain volume. A normalized cosine overlap is dimensionless. A risk under squared loss has squared-target units, not target units.

A useful record for each operation contains the parent ids, domain, field units, probe or kernel version, channel order, coefficients, covariance convention, background, retained residual and boundary rule. It also contains the operation type: exact on stored arrays, approximate after remapping, fitted operator, or proposed physical mechanism. A reader should be able to reconstruct the result from that record without inferring what a color or a pocket label meant.

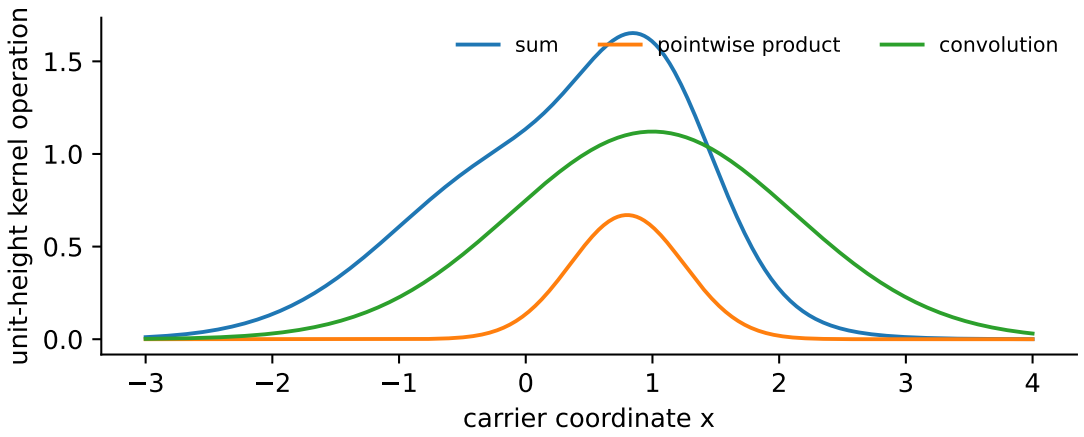


Figure 1: Exact operations on the two unit-height kernels in the Gaussian worked example. The sum, product and convolution have different centers, widths and amplitudes. This is an analytical illustration.

For downstream prediction, distinguish current represented arithmetic from future-input uncertainty. If $V_C = V_A + V_B + \gamma V_A V_B$, a perfectly known γ still leaves uncertainty in future V_A, V_B. Linearizing at predicted means gives

$$\delta V_C \simeq (1 + \gamma V_B)\delta V_A + (1 + \gamma V_A)\delta V_B + V_A V_B \delta\gamma. \tag{13.1}$$

Taking the covariance of this expression propagates cross-dependence. A first-order approximation can fail for large errors, strong nonlinearity or branch changes; then transform joint samples or use the full probabilistic model.

14 Exercises and solution trails

Exercise 14.1. Let $U_A = 4$, $U_B = 9$. Find the admissible range of U_{A+B} and compute it when $\Gamma = -3$.

Solution. The bounds are $(2-3)^2 = 1$ and $(2+3)^2 = 25$. The specified cross term gives $4 + 9 - 6 = 7$. The two energies alone leave a continuum of possibilities.

Exercise 14.2. Let $V - b = (2, 4)$ and ownership masks $\chi_1 = (1/2, 1/4)$, $\chi_2 = (1/4, 1/4)$. Recover the residual and check reconstruction.

Solution. $V_1 = (1, 1)$, $V_2 = (1/2, 1)$ and $R = (1/2, 2)$. Their sum is $(2, 4)$. Omitting R would change the carrier.

Exercise 14.3. For design columns $z_1 = (1, 2)$ and $z_2 = (2, 4)$, explain why a unique ridge solution does not certify two recovered interaction coefficients.

Solution. $Z(-2, 1)^\top = 0$. The data determine only $\theta_1 + 2\theta_2$. Ridge selects one point on this equivalence line using its penalty. A new noncollinear observation is required to identify both coefficients without that restriction.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. This paper and its companion (Papers 1 and 2 of the series) share the archival DOI 10.5281/zenodo.23781662.

References

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